

MRI COIL TUNING & MATCHING – GETTING IT RIGHT

Tuning (Resonance) + Matching (Impedance) deliver maximum RF power to the patient and maximum signal from the patient.

GOAL: Maximize Power Transfer (TX) and Signal Transfer (RX) → High SNR → Better Images

1. WHY TUNING & MATCHING MATTER

GOOD TUNING & MATCHING

- ✓ Max power to patient (TX)
- ✓ Max signal from patient (RX)
- ✓ High SNR and image quality
- ✓ Uniform flip angle (TX)
- ✓ System runs cool and stable

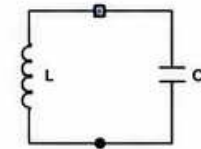
POOR TUNING OR MATCHING

- ✗ Power reflected back (TX)
- ✗ Signal lost in coil (RX)
- ✗ Low SNR, poor images
- ✗ Uneven flip angle
- ✗ Overheating, possible trips

Tune to the right frequency. Match to the right impedance.
Every time.

2. TUNING (RESONANCE)

Tuning sets the coil to resonate at the Larmor frequency (f_0).



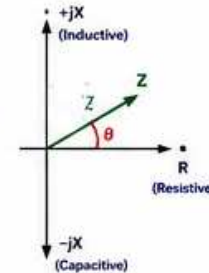
$$\text{Resonant Frequency } f_0 = \frac{1}{2\pi\sqrt{LC}}$$

At f_0 : $X_L = X_C \rightarrow$ Net Reactance (X) = 0 Ω
(coil is purely resistive)

3. IMPEDANCE BASICS

Impedance $Z = R + jX$

R = Resistance (Ω)
(energy loss)
X = Reactance (Ω)
(stored energy)



At resonance: $X = 0 \rightarrow Z$ is on the real axis (pure R).

4. MATCHING (IMPEDANCE)

Matching transforms the coil impedance (Z_{coil}) to the system reference impedance (Z_{sys}) (typically 50 Ω).

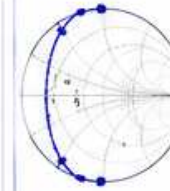


Best power/signal transfer when:

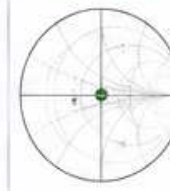
Z_{coil} (after matching) = Z_{sys}^* (conjugate match)
For 50 Ω system with real load: $Z = 50 + j0 \Omega$

5. IMPEDANCE ON SMITH CHART (50 Ω SYSTEM)

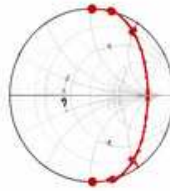
Inductive (+jX)



Good Match (50 + j0 Ω)



Capacitive (-jX)



Center = 50 + j0 Ω (Perfect Match)
Move toward center to improve match.

6. TYPICAL TUNING & MATCHING WORKFLOW

- 1 Connect coil to analyzer.
- 2 Tune to f_0 (resonance):
Adjust variable capacitors so $X \approx 0$ (minimum $|X|$).
- 3 Match to 50 Ω :
Adjust matching capacitors for $Z = 50 + j0 \Omega$.
- 4 Verify:
Check Return Loss (RL) or VSWR.
- 5 Record settings:
Document capacitor positions.
- 6 Recheck after changes:
Cable, loading, or repairs.

PRO TIP
Always tune FIRST, then match.
Re-check tuning after matching.



7. KEY MEASUREMENTS (WHAT THEY TELL YOU)

A. RETURN LOSS (RL)

How much power is reflected back.

$$RL = -20 \log_{10} (|\Gamma|) \text{ dB}$$

$$\text{Where } \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$Z_0 = 50 \Omega$ (system)

$Z_L =$ load (coil after matching)

RL (dB)	Reflected Power
> 20 dB	< 1 %
15 dB	3.2 %
10 dB	10 %
6 dB	25 %

Goal: RL $\geq$ 20 dB (normal)
Minimum acceptable: RL $\geq$ 10 dB

B. VSWR

Voltage Standing Wave Ratio

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

VSWR	Match Quality
1.0 : 1	Perfect
1.5 : 1	Good
2.0 : 1	Fair
3.0 : 1	Poor
∞ : 1	Open/Short

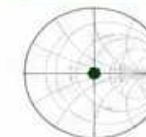
Goal: VSWR $\leq$ 1.5 : 1
Maximum acceptable: VSWR $\leq$ 2 : 1

C. IMPEDANCE (Z)

Analyzer reads $Z = R + jX$

- At resonance: $X \approx 0$
- After matching: $Z \approx 50 + j0 \Omega$

What to look for:
• R close to 50 Ω
• X close to 0 Ω



D. Q (QUALITY FACTOR)

$$Q = \frac{f_0}{\text{Bandwidth}}$$

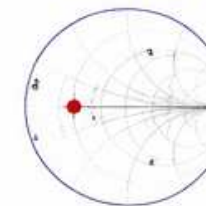
- High Q = narrow bandwidth (good for Rx sensitivity)
- Lower Q = wider bandwidth (good for Tx efficiency)

Typical Q (receive coils):
50 – 150 (varies by coil type)

8. EXAMPLE: BEFORE & AFTER MATCHING

BEFORE MATCHING
(Example)

$Z = 15 + j40 \Omega$

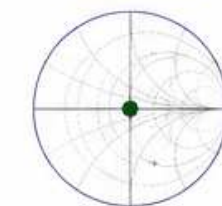


$|\Gamma| = 0.72$ RL = 2.8 dB
VSWR = 6.1 : 1

Poor match – high reflection

AFTER MATCHING
(Good)

$Z = 50 + j1 \Omega$



$|\Gamma| = 0.045$ RL = 26.9 dB
VSWR = 1.09 : 1

Good match – low reflection

9. COMMON CAUSES OF POOR TUNE/MATCH

- Wrong capacitor settings
- Coil detuned (during TX) / not tuned (RX)
- Cable length or position change
- Poor or loose connections
- Coil loading change (patient position/size)
- Failed capacitor or component
- Incorrect matching network configuration

10. TROUBLESHOOTING GUIDE

Symptom	Possible Cause	What to Check / Do
High Reflected Power (Low RL)	Poor match	Adjust matching capacitors
High Reactance (X)	Not at resonance	Retune coil to f_0
Good Tune, Poor Match	Wrong match settings	Adjust matching capacitors
Values drift	Loose connections / heat	Check connectors, re-tighten
System Trips / Overheat	High reflected power	Check RL, fix match
Low SNR	Poor match (Rx)	Optimize match for Rx

11. PRACTICAL TIPS

- ✓ Use a calibrated network analyzer.
- ✓ Use short, high-quality coax cables.
- ✓ Always tune with coil unloaded, then recheck with loading.
- ✓ Mark capacitor positions when optimum.
- ✓ Re-check after moving cables or coil.
- ✓ Keep coils and components clean and dry.
- ✓ Document final settings and measurements.

12. TARGET VALUES (QUICK REFERENCE)

Parameter	Target (50 Ω System)	Acceptable
Resonance (X)	$X \approx 0 \Omega$ at f_0	$ X < 2 \Omega$
Impedance (Z)	50 + j0 Ω	R: 45–55 Ω , $ X < 5 \Omega$
Return Loss (RL)	≥ 20 dB	≥ 10 dB
VSWR	$\leq 1.5 : 1$	$\leq 2 : 1$
Reflected Power	≤ 1 %	≤ 10 %

13. QUICK CHECK FLOW

- 1 Tune to f_0 ($X \approx 0$)
- 2 Match to 50 + j0 Ω
- 3 Check RL ≥ 20 dB
- 4 Check VSWR $\leq 1.5 : 1$
- 5 Verify with loading
- 6 Document & done



ENGINEER NOTES

- Tuning = Resonance ($X \rightarrow 0$)
- Matching = Impedance ($Z \rightarrow 50 + j0 \Omega$)
- Good tune + good match = Max power & max signal

ABBREVIATIONS

f_0 = Larmor Frequency (MHz)
Z = Impedance (Ω)
R = Resistance (Ω)
X = Reactance (Ω)
RL = Return Loss (dB)
VSWR = Voltage Standing Wave Ratio
Q = Quality Factor



SAFETY REMINDER

- High RF power can cause tissue heating.
- Always follow site safety procedures.
- Verify ESD precautions when handling components.



THE BOTTOM LINE

Tune to the right frequency.
Match to the right impedance.
Deliver power in. Get signal out.
That's how we get the best MR images.